

8(a)	<i>any five from:</i> 1 mark–release–recapture ; 2 capture (sample of) rats and count them ; 3 <i>ref. to humane traps</i> ; 4 mark so as to not adversely affect rats or describe method of marking such as, tagging / using dyes ; 5 return to where they were caught / AW ; 6 allow time to mix (with population) / AW ; 7 capture second sample ; 8 count number caught and number marked ; 9 population size estimate (M_1) = $\frac{\text{number in first sample } (N_1) \times \text{number in second sample } (N_2)}{\text{number in second sample that are marked } (M_2)}$; 10 AVP ; e.g. repeat	5
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Question	Answer	Marks
8(b)(i)	$r_s = 1 - \left(\frac{6 \times 164}{8^3 - 8} \right)$ or $= 1 - \left(\frac{984}{504} \right)$ or $= 1 - 1.95 ;$ $r_s = -0.95 ;$	2
8(b)(ii)	there is a strong correlation ; negative correlation / as the altitude increases the mean height of the soft rush decreases ;	2